BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Programming: Unseen Problems and Debugging

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does WHERE do in an SQL query?

2. When does a compiler translate source code?
   A) When a key is pressed
   B) Only after the program ends
   C) Before the program runs
   D) Only while each line runs

3. "A router's main job is executing spreadsheet formulas." Correct this. Explain why.

4. A computer sends data to a printer on its LAN. "A switch sends every unicast frame to every port." Correct this. How should the data reach the printer?

Today's objectives

Explain the core ideas of programming: unseen problems and debugging using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Understand the scenario
Identify input, processing, output and restrictions before coding.

Write a plan
Use simple steps, selections and loops before translating into code.

Trace unseen code
Record value changes and follow each condition accurately.

Debug and improve
Compare expected and actual behaviour, correct the cause and retest.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Understand the scenario
Identify input, processing, output and restrictions before coding.

Write a plan
Use simple steps, selections and loops before translating into code.

Trace unseen code
Record value changes and follow each condition accurately.

Debug and improve
Compare expected and actual behaviour, correct the cause and retest.

I Do - Worked model

An unknown program calculates price = quantity * unitPrice but prints unitPrice. Find the fault.

- Identify expected output as total price.
- Trace sample quantity=3, unitPrice=4; price=12.
- The print statement uses the wrong variable.
- Change to print(price), retest output 12.

We Do - Guided practice

1. A program should count 1 to 5 but stops at 4. Explain a fix.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the operator in a storage cost-per-GB calculation.

```
price = 48
capacityGB = 12
costPerGB = price * capacityGB
print(costPerGB)
```

You Do - Independent exam questions

Q2. Explain why a trace is useful for unfamiliar code. [3 marks]

Q3. Correct an algorithm that checks 1<=x<=10 using OR between two comparisons. [2 marks]

Q4. State two steps to take after correcting a logic error. [2 marks]

Exit Ticket - True or False?

A trace can help locate the first incorrect variable update.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It selects rows that meet a condition.

2. C - Before the program runs

3. It forwards traffic between different networks using IP addressing.

4. A switch uses MAC addresses to send data to the right connected device. It does not normally send each known unicast frame to every port.

I Do - Explain the model

Identify expected output as total price.

Trace sample quantity=3, unitPrice=4; price=12.

The print statement uses the wrong variable.

Change to print(price), retest output 12.

We Do - Model answers

1. Test both first and last values.

Code First - Answer

price = 48
capacityGB = 12
costPerGB = price / capacityGB
print(costPerGB)
Expected output: 4

You Do - Self-mark

Q2: It makes executed steps and changing variable values explicit, exposing incorrect assumptions or logic.

Q3: Use x>=1 AND x<=10.

Q4: Run/test with expected normal and boundary cases, and compare outcomes.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.